

Trainings ▾

General Information

The aftertreatment diesel particulate filter (DPF) differential pressure sensor is a combination sensor that monitors the differential pressure across the DPF, as well as the pressure at the outlet of the DPF.

The aftertreatment diesel particulate filter differential pressure sensor is located on the aftertreatment diesel particulate filter.

Initial Check

Use an electronic service tool to monitor the value of the aftertreatment diesel particulate filter differential pressure sensor with the key in the ON position and the engine off.

The value of the aftertreatment diesel particulate filter differential pressure should meet the following specification with the key in the ON position and the engine off.

Aftertreatment Diesel Particulate Filter Differential Pressure

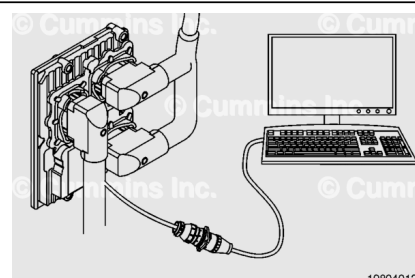
kpa		in-hg
0 ± 3	NOM	0 ± 0.89

The value of the aftertreatment diesel particulate filter outlet pressure sensor should meet the following specification with the key in the ON position and the engine off.

Aftertreatment Diesel Particulate Filter Outlet Pressure

kpa		in-hg
0 ± 3	NOM	0 ± 0.89

If either value is out of specification, inspect the aftertreatment diesel particulate filter differential pressure tubes for blockage.



If the aftertreatment diesel particulate filter differential pressure tubes are **not** blocked, and if either sensor reading is out of specification, replace the aftertreatment diesel particulate filter differential pressure sensor.

The value of the aftertreatment diesel particulate filter outlet pressure sensor **must** meet the following specification with the key in the ON position and the engine off.

Aftertreatment Diesel Particulate Filter Outlet Pressure Sensor Voltage (VDC)	
Min	Max
0.5	4.5

Replace the aftertreatment diesel particulate filter differential pressure sensor if the voltage recorded does **not** fall within the specified voltage range.

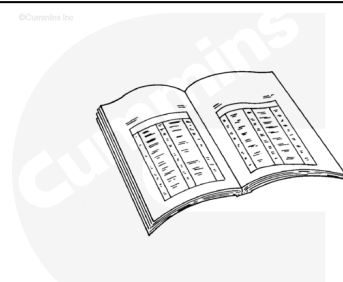
Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the OEM service manual.

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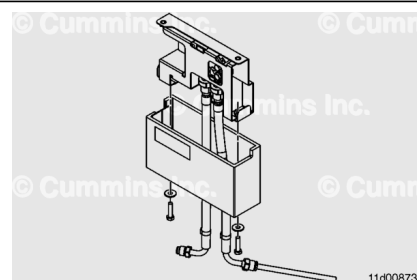


ck800wa

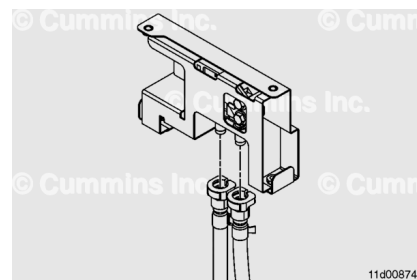
Remove

If equipped with a remote mounted aftertreatment DPF differential pressure sensor:

- Remove the capscrews that secure the aftertreatment DPF differential pressure sensor insulation box
- Slide the aftertreatment DPF differential pressure sensor insulation box down the aftertreatment DPF differential pressure sensor tubes.

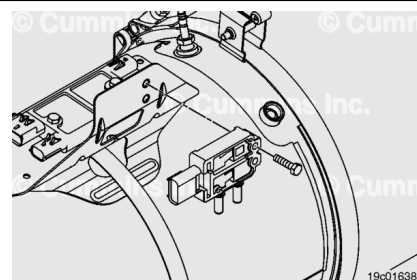


Remove the quick disconnect fittings of the differential pressure sensor tubes from the aftertreatment differential pressure sensor by pressing in the locking tangs on the quick disconnect fitting.



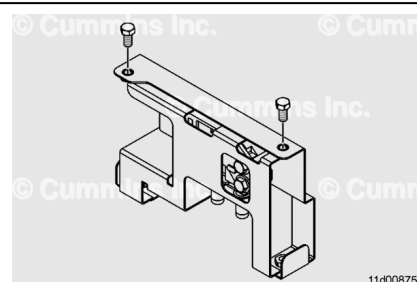
If equipped with a DPF mounted aftertreatment DPF differential pressure sensor:

- Disconnect the sensor connector from the wiring harness
- Remove the mounting capscrew.



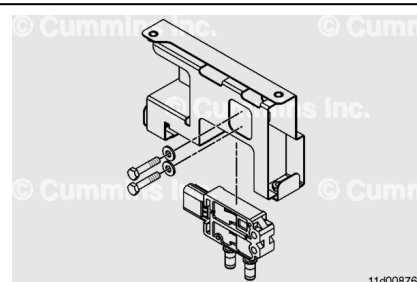
If equipped with a remote mounted aftertreatment DPF differential pressure sensor:

- Depending on the OEM installation it may be easier to access the aftertreatment DPF differential pressure sensor if the aftertreatment DPF differential pressure sensor insulation box frame is removed.
- Remove the fasteners that secure the aftertreatment DPF differential pressure sensor insulation box frame. Refer to the OEM service manual.



Disconnect the sensor connector from the wiring harness.

Remove the capscrews that secure the aftertreatment DPF differential pressure sensor.



Clean and Inspect for Reuse

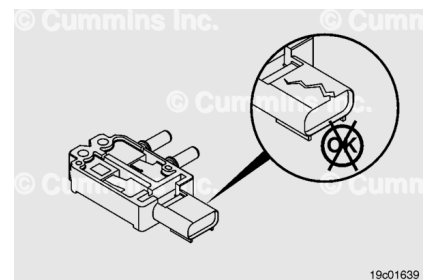
Inspect the inside of the tube connections on the sensor for plugging or soot accumulation. If plugging or soot accumulation is observed, the aftertreatment diesel particulate

filter differential pressure tubes **must** be cleaned.

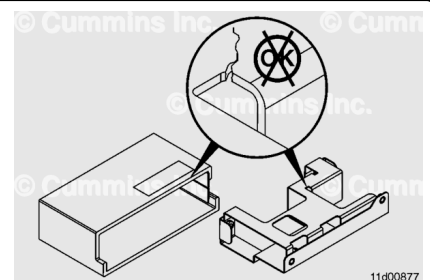
Inspect the engine harness connector and sensor for the following:

- Cracked or broken connector shell
- Missing or damaged connector seals
- Dirt, debris, or moisture in or on the connector pins
- Corroded, bent, broken, pushed back, or expanded pins
- Chipped, cracked, extruded, or damaged sensor.

Repair or replace parts as necessary.



If equipped with a remote mounted aftertreatment DPF differential pressure sensor, inspect the aftertreatment DPF differential pressure sensor insulation box and cage for damage. Replace if damage is found.



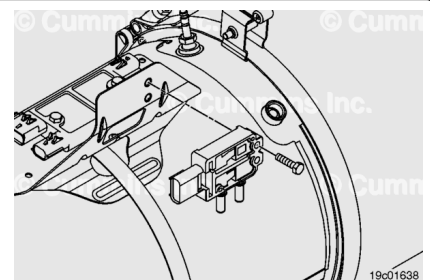
Install

If equipped with a DPF mounted aftertreatment DPF differential pressure sensor, install the aftertreatment DPF differential pressure sensor onto the mounting bracket.

Install and tighten the mounting capscrew.

Torque Value: 14 n•m [124 in-lb]

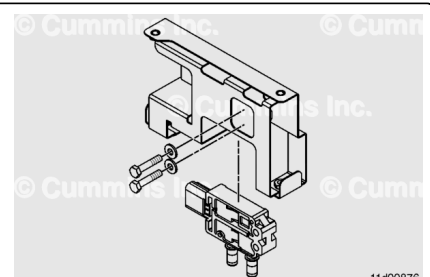
Connect the wiring harness to the aftertreatment DPF differential pressure sensor.



If equipped with a remote mounted aftertreatment DPF differential pressure sensor, and if removed install the aftertreatment diesel particulate filter differential pressure sensor insulation box cage onto the chassis. Refer to the OEM service manual.

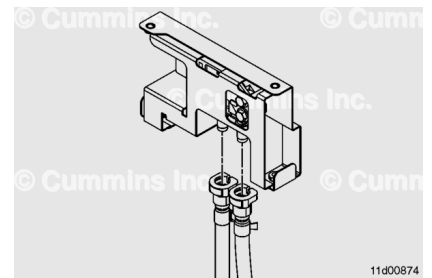
Install the aftertreatment diesel particulate filter differential pressure sensor onto the mounting bracket. Install and tighten the mounting capscrews.

Torque Value: 14 n•m [124 in-lb]



Connect the wiring harness to the aftertreatment DPF differential pressure sensor.

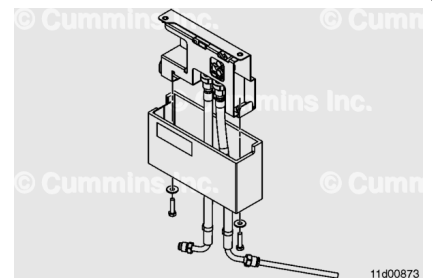
Install the quick disconnect fittings of the differential pressure sensor tubes to the aftertreatment differential pressure sensor.



Slide the aftertreatment DPF differential pressure sensor insulation box up the aftertreatment DPF differential pressure sensor tubes.

Install the capscrews that secure the aftertreatment DPF differential pressure sensor insulation box.

Torque Value: 14 N·m [124 in-lb]



Finishing Steps

⚠ WARNING ⚠

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- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.